Name : Sanjana Kiran More
Class : MCA Part-II Sem-III
Subject : Data Analytics

Date: / / 2026
Roll No. :27

Program 1

Code:

```
import numpy as np

arr = np.array(["Hello World", "Python Programming", "NumPy Library"])

result = [i.split() for i in arr]

print("Original Array:")

print(arr)

print("\nArray after splitting elements:")

print(result)
```

Output:

```
Original Array:
['Hello World' 'Python Programming' 'NumPy Library']

Array after splitting elements:
[['Hello', 'World'], ['Python', 'Programming'], ['NumPy', 'Library']]
```

Program 2

Code:

```
import numpy as np
arr = np.array([2, 4, 5, 2, 6, 2, 4, 5, 2, 7, 5, 5])
values, counts = np.unique(arr, return_counts=True)
most_frequent = values[np.argmax(counts)]
frequency = counts[np.argmax(counts)]
print("Array:", arr)
print("Most Frequent Value:", most_frequent)
print("Frequency:", frequency)
```

Output:

```
Array: [2 4 5 2 6 2 4 5 2 7 5 5]
Most Frequent Value: 2
Frequency: 4
```

Program 3

Code:

```
import numpy as np
random_numbers = np.random.normal(loc=0, scale=1, size=5)
print("Five random numbers from normal distribution:")
print(random_numbers)
```

Output:

```
Five random numbers from normal distribution:
[-1.290475   -0.69507079  1.66506563 -1.41980574  0.90592776]
```

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Program 4

Code:

```
import numpy as np

arr = np.array([9, 4, 7, 2, 5, 8, 1])

n = 4

arr[:n] = np.sort(arr[:n])

print("Updated Array:")

print(arr)
```

Output:

```
Updated Array:
[2 4 7 9 5 8 1]
```

Program 5

Code:

```
import numpy as np
arr = np.array([12, 45, 7, 23, 89, 34, 56])
maximum = np.max(arr)
minimum = np.min(arr)
difference = maximum - minimum
print("Array:", arr)
print("Maximum Value:", maximum)
print("Minimum Value:", minimum)
print("Difference:", difference)
```

Output:

```
Array: [12 45  7 23 89 34 56]
Maximum Value: 89
Minimum Value: 7
Difference: 82
```

Program 6

Code:

```
import numpy as np
vector1 = np.array([2, 3, 4])
vector2 = np.array([5, 6, 7])
cross_product = np.cross(vector1, vector2)
print("Vector 1:", vector1)
print("Vector 2:", vector2)
print("Cross Product:", cross_product)
```

Output:

```
Vector 1: [2 3 4]
Vector 2: [5 6 7]
Cross Product: [-3  6 -3]
```

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Program 7

Code:

```
import numpy as np
array_3d = np.array([
[[1, 2, 3], [4, 5, 6]],
[[7, 8, 9], [10, 11, 12]]
])
print("Three-Dimensional Array:")
print(array_3d)
print("Shape:", array_3d.shape)
```

Output:

```
Three-Dimensional Array:
[[[ 1  2  3]
  [ 4  5  6]]

 [[ 7  8  9]
  [10 11 12]]]
Shape: (2, 2, 3)
```

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Program 8

Code:

```
import numpy as np

import pandas as pd

arr = np.array([10, 20, 30, 40, 50])

series = pd.Series(arr)

print("NumPy Array:")

print(arr)

print("\nPandas Series:")

print(series)
```

Output:

```
NumPy Array:
[10 20 30 40 50]

Pandas Series:
0    10
1    20
2    30
3    40
4    50
dtype: int64
```

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Program 9

Code:

```
import pandas as pd
df = pd.DataFrame({
'Name': ['Amit', 'Riya', 'Rahul', 'Sneha'], 'Age': [21, 22, 20, 23],
'Marks': [85, 90, 88, 92]
})
series = df.iloc[:, 0]
print("Original DataFrame:")
print(df)
print("\nFirst Column as Series:")
print(series)
```

Output:

```
Original DataFrame:
    Name  Age  Marks
0   Amit   21     85
1   Riya   22     90
2  Rahul   20     88
3  Sneha   23     92

First Column as Series:
0     Amit
1     Riya
2    Rahul
3    Sneha
Name: Name, dtype: str
```

Program 10

Code:

```
import pandas as pd
df1 = pd.DataFrame({
'Name': ['Amit', 'Riya'], 'Age': [21, 22]})
df2 = pd.DataFrame({
'Name': ['Rahul', 'Sneha'], 'Age': [20, 23] })
result = pd.concat([df1, df2], ignore_index=True)
print("First DataFrame:")
print(df1)
print("\nSecond DataFrame:")
print(df2)
print("\nCombined DataFrame:")
print(result)
```

Output:

```
First DataFrame:
   Name  Age
0  Amit   21
1  Riya   22

Second DataFrame:
    Name  Age
0  Rahul   20
1  Sneha   23

Combined DataFrame:
    Name  Age
0   Amit   21
1   Riya   22
2  Rahul   20
3  Sneha   23
```

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Program 11

Code:

```
import pandas as pd
df = pd.DataFrame({
'Name': ['Amit', 'Riya', 'Rahul', 'Sneha', 'Vijay'], 'Age': [21, 22, 20, 23, 24],
'Marks': [85, 90, 88, 92, 80]
})
df1 = df.iloc[:3]
df2 = df.iloc[3:]
print("Original DataFrame:")
print(df)
print("\nFirst DataFrame:")
print(df1)
print("\nSecond DataFrame:")
print(df2)
```

Output:

```
Original DataFrame:
    Name  Age  Marks
0   Amit   21     85
1   Riya   22     90
2  Rahul   20     88
3  Sneha   23     92
4  Vijay   24     80

First DataFrame:
    Name  Age  Marks
0   Amit   21     85
1   Riya   22     90
2  Rahul   20     88

Second DataFrame:
    Name  Age  Marks
3  Sneha   23     92
4  Vijay   24     80
```

Program 12

Code:

```
import pandas as pd
df = pd.read_excel("D:\\practicals/student.xlsx")
print("Excel Data:")
print(df)
```

Output:

```
Excel Data:
   Roll_No First_Name Last_Name Course  Sem
0      101       Riya    Rathod    MBA    2
1      102     Shreya   Gaikwad  Btech    4
2      103     Saheba   Dilawar    BBA    1
3      104    Akansha    Phalke    BCA    3
4      105    Sanjana      More    MCA    2
```

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Program 13

Code:

```
import pandas as pd
import numpy as np
df = pd.DataFrame({
'Name': ['Amit', 'Riya', 'Rahul', 'Sneha'], 'Marks': [85, np.nan, 90, np.nan],
'Age': [21, 22, np.nan, 23]
})
print("Original DataFrame:")
print(df)
print("\nMissing Values:")
print(df.isnull())
df = df.fillna(0)
print("\nDataFrame after replacing missing values:")
print(df)
```

Output:

```
Original DataFrame:
    Name  Marks   Age
0   Amit   85.0  21.0
1   Riya    NaN  22.0
2  Rahul   90.0   NaN
3  Sneha    NaN  23.0

Missing Values:
    Name  Marks    Age
0  False  False  False
1  False   True  False
2  False  False   True
3  False   True  False

DataFrame after replacing missing values:
    Name  Marks   Age
0   Amit   85.0  21.0
1   Riya    0.0  22.0
2  Rahul   90.0   0.0
3  Sneha    0.0  23.0
```

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Program 14

Code:

```
import pandas as pd
df = pd.DataFrame({
'Department': ['IT', 'HR', 'IT', 'Finance', 'HR'],
'Employee': ['Amit', 'Riya', 'Rahul', 'Sneha', 'Vijay'], 'Salary': [50000, 45000, 55000, 60000,
48000]
})
grouped = df.groupby('Department')
for name, group in grouped:
    print("Department:", name)
    print(group)
    print()
```

Output:

```
Department: Finance
  Department Employee  Salary
3    Finance    Sneha   60000

Department: HR
  Department Employee  Salary
1         HR     Riya   45000
4         HR    Vijay   48000

Department: IT
  Department Employee  Salary
0         IT     Amit   50000
2         IT    Rahul   55000
```

Program 15

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
df = pd.DataFrame({
'Month': ['Jan', 'Feb', 'Mar', 'Apr', 'May'], 'Sales': [200, 300, 250, 400, 350]
})
df.plot(x='Month', y='Sales', kind='line', marker='o')
plt.title("Monthly Sales Report")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.show()
```

Output: